

SOFTWARE ENGINEERING ASSIGNMENT

EMERGENCY AMBULANCE MANAGEMENT SYSTEM

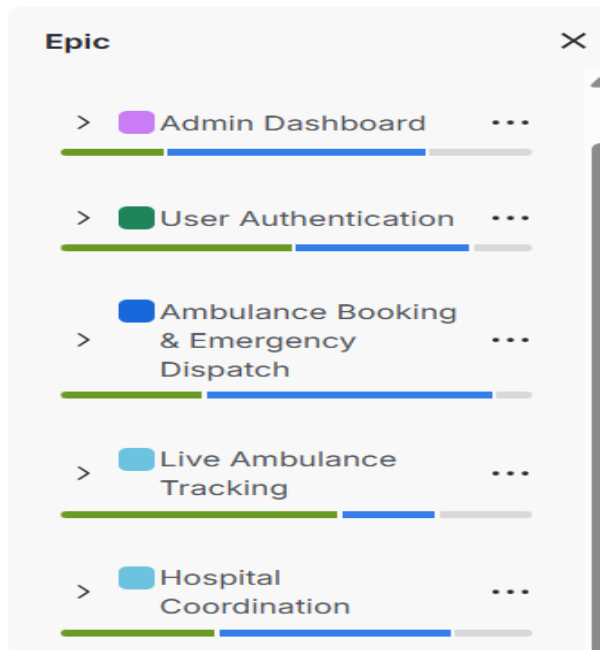
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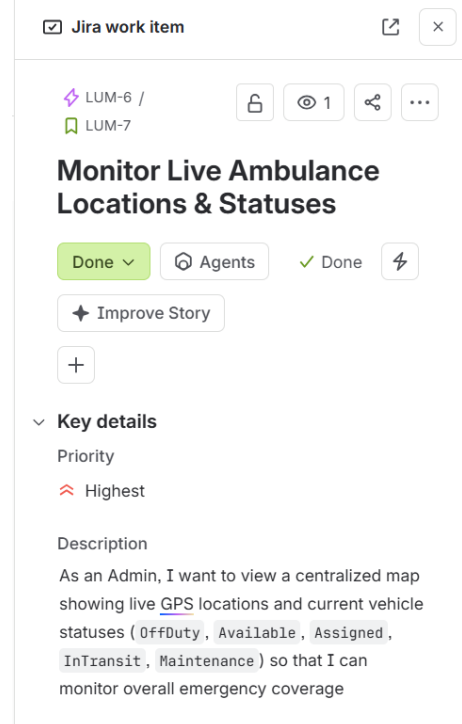
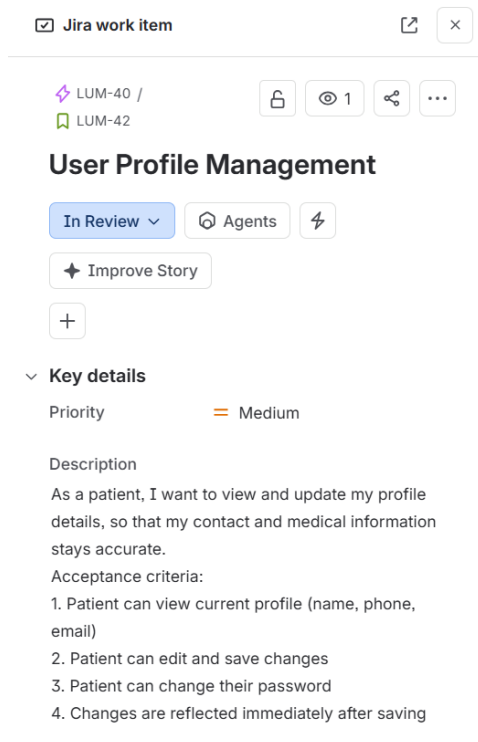
WHAT IS EMERGENCY AMBULANCE MANAGEMENT SYSTEM?

- The Emergency Ambulance Management System is a digital platform developed to make the process of booking and managing ambulances easier during emergency situations. Instead of depending on manual calls and coordination, the system allows patients to request an ambulance, share their location, and track the ambulance until it reaches them.
- The system has different users such as Patients, Ambulance Drivers, and Hospital Staff. Patients can book an ambulance, select the required ambulance type, provide their pickup location, and view the ambulance's current location and estimated arrival time. Ambulance drivers can receive booking requests, navigate to the pickup location, and coordinate with hospitals.
- The Hospital Coordination module helps find suitable hospitals, check their availability, notify hospitals about incoming ambulances, and share important patient information. The system also provides emergency notifications so that the concerned users can receive updates about the ambulance and hospital coordination.
- The major modules of the project are Ambulance Booking, Live Ambulance Tracking, Hospital Coordination, Emergency Profile Management, and Emergency Notifications & Communication. The project is managed using Agile Scrum in Jira, where the team creates Epics, User Stories, Tasks/Subtasks, and Sprints to organize and track the project work.

EPIC PANEL:



USER STORIES:



Jira work item

LUM-48 / LUM-53

Auto-Detect and Select Pickup Location

To Do

Agents

⚡

Improve Story

+

Key details

Priority

High

Description

User story

As a patient, when I need an ambulance, I want the system to auto-detect my GPS location or let me set a pin on the map so that the ambulance arrives at the exact spot.

Context

The issue is about improving pickup location selection for ambulance arrival accuracy.

Acceptance criteria

- The system can auto-detect the patient's GPS location.
- The patient can set a pin on the map manually.
- The selected location helps the ambulance arrive at the exact spot.

Jira work item

LUM-64 / LUM-66

Display Ambulance Route to Pickup Location

Done

Agents

✓ Done

⚡

Improve Story

+

Key details

Priority

High

Description

As a Patient, I want to see the ambulance's route from its current location to my pickup point so that I can understand the expected path and progress of the ambulance.

SUB TASKS:

Subtasks

100% Done

Work	Pr...	A...	Status
LUM-16 Design UI layout for central GPS map view	= N	AR	Done
LUM-17 Create API endpoints to fetch live ambulance locations and...	= N	AR	Done
LUM-18 Implement real-time map pin updates using WebSockets	= N	AR	Done
LUM-127 Configure WebSocket API for real-time GPS streaming	= N	AR	Done
LUM-128 Build interactive map UI component for vehicle pins	= N	AR	Done

Subtasks

...  +

Work	Pr...	A...	Status
 LUM-80 Integrate Google Maps	= N	N	Done 
 LUM-81 Display ambulance-to-pickup route	= N	N	In Progress 
 LUM-82 Update route based on ambulance location	= N	N	To Do 

33% Done

Subtasks

...  +

Work	Pr...	A...	Status
 LUM-62 Integrate Google Maps Geolocation API for pickup...	= N	C	To Do 

0% Done

PRODUCT BACKLOG:

☐ Backlog (3 work items)	0 0 0  Create sprint
☒ LUM-2 Ambulance Booking	 Ambulance Bookin... In Progress  - 
☒ LUM-3 Live Ambulance Tracking	 Live Ambulance Tr... To Do  - 
☒ LUM-4 Hospital Coordination	 Hospital Coordinat... In Progress  - 
+ Create	

SPRINT BACKLOG:

☐ LUM Sprint 1 26 Aug – 9 Sep (36 work items)	11 47 49 Complete sprint ...
 LUM-1 User Registration	  Done  4  VT
 LUM-41 User Login	  Done  4  VT
 LUM-42 User Profile Management	  In Review  4 = VT
 LUM-104 User Logout	  Done  - > VT
 LUM-105 Forgot / Reset Password	  Done  - = VT
 LUM-106 Email/Phone Verification (OTP)	  In Progress  - = VT
 LUM-107 Account Deactivation	  In Review  - > VT
 LUM-108 Role-Based Access Control	  To Do  - < VT
 LUM-7 Monitor Live Ambulance Locations & Statuses	  Done  5 >> AR
 LUM-8 Override & Reassign Emergency Dispatches	  In Progress  5 < AR
 LUM-9 Manage Active Driver Shifts & Statuses	  In Progress  3 = AR
 LUM-10 Track Real-Time Hospital Bed Capacity	  In Progress  2 < AR

LUM Sprint 2 23 Aug – 26 Aug (10 work items) 0 25 10 Complete sprint ...					
LUM-42	User Profile Management	User Authentication	In Review	4	VT
LUM-104	User Logout	User Authentication	Done	-	VT
LUM-10	Track Real-Time Hospital Bed Capacity	Admin Dashboard	In Progress	3	AR
LUM-12	View Emergency System Analytics & Response Metrics	Admin Dashboard	In Progress	5	AR
LUM-57	Select Ambulance Equipment Level	Ambulance Bookin...	In Review	3	C
LUM-142	Select Destination Hospital or Medical Center	Ambulance Bookin...	In Progress	3	C
LUM-145	Cancel Emergency Booking with Reason	Ambulance Bookin...	In Review	2	C
LUM-65	View Live Ambulance Location	Live Ambulance Tr...	Done	5	N
LUM-71	Check Hospital Availability	Hospital Coordinat...	In Progress	5	N
LUM-68	Update Ambulance Booking Status in Real Time	Live Ambulance Tr...	Done	5	N
+ Create					

LUM Sprint 3 27 Aug – 29 Aug (9 work items) 6 7 5 Complete sprint ...					
LUM-105	Forgot / Reset Password	User Authentication	Done	-	VT
LUM-106	Email/Phone Verification (OTP)	User Authentication	In Progress	-	VT
LUM-13	Receive Automated Delay & Dispatch Alerts	Admin Dashboard	To Do	3	AR
LUM-150	Access Booking History and Download Medical Transport Invoices	Ambulance Bookin...	In Progress	-	C
☐ LUM-151	Direct Phone Call with Assigned Ambulance Driver via Masked Number	Ambulance Bookin...	In Progress	-	C
LUM-164	Post-Trip Driver & Service Rating	Ambulance Bookin...	In Review	2	C
LUM-72	Notify Hospital About Incoming Ambulance	Hospital Coordinat...	Done	5	N
LUM-73	Share Patient Information With Hospital	Hospital Coordinat...	In Progress	5	N
LUM-74	Confirm Hospital Acceptance	Hospital Coordinat...	To Do	3	N

SCRUM BOARD:

Spaces

Lumos

...

Summary

List

Board

Backlog

Development

Forms

Timeline

Docs

Reports

Q Search board

N

AR

C

VT

Filter

Group

Complete sprint

To Do 5

Role-Based Access Control

Parent

User Authentication

LUM-108

Subtasks 1/4

Receive Automated Delay & Dispatch Alerts

Parent

admin alerts notifications

Admin Dashboard

LUM-13

Subtasks 0/5

+ Create

In Progress 11

Email/Phone Verification (OTP)

Parent

User Authentication

LUM-106

Subtasks 2/4

Track Real-Time Hospital Bed Capacity

Parent

admin bed-availability hospital_coordination

Admin Dashboard

LUM-10

Subtasks 3

+ Create

In Review 5

User Profile Management

Parent

auth

User Authentication

LUM-42

Subtasks 1/4

Account Deactivation

Parent

User Authentication

LUM-107

Subtasks 1/3

Select Ambulance Equipment

+ Create

Done 7

User Logout

Parent

User Authentication

LUM-104

Subtasks 3/3

Forgot / Reset Password

Parent

User Authentication

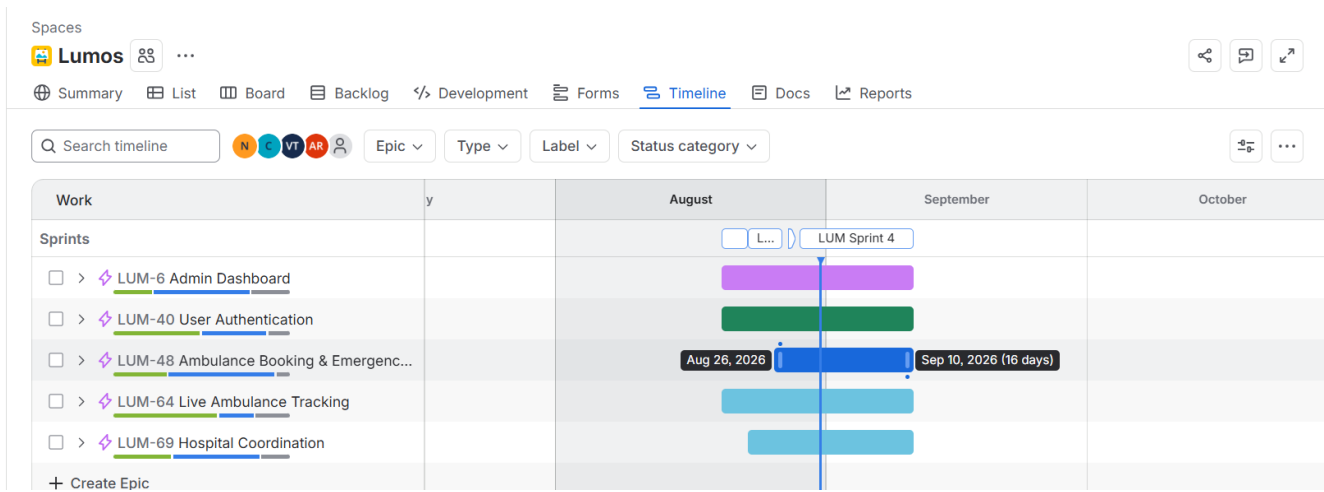
LUM-105

Subtasks 2/4

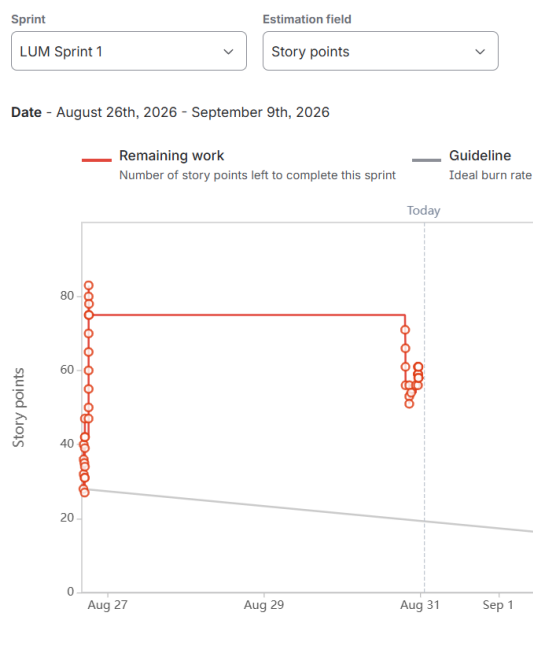
View Pre-Arrival Emergency First-Aid Instructions

+ Create

EPIC TIMELINE:



BURNDOWN CHART:



SPRINT RETROSPECTIVE – EMERGENCY AMBULANCE MANAGEMENT SYSTEM

1. Sprint Overview

In this sprint, our team worked on the Emergency Ambulance Management System using the Agile Scrum methodology in Jira. We divided the project into different Epics based on the main features of the system, such as Ambulance Booking, Live Ambulance Tracking, Hospital Coordination, Emergency Profile Management, and Emergency Notifications.

For Sprint 1, we mainly focused on the important parts of the emergency process, starting from booking an ambulance to tracking it and coordinating with the hospital. We created User Stories based on the requirements and further divided them into smaller Subtasks. The features that could not be completed in this sprint were kept in the Product Backlog for the next sprint.

2. What Went Well

- We were able to divide the project into different Epics and User Stories, which made the work easier to organize.
- The User Stories were written in the Scrum format, such as "As a Patient, I want to..., so that..."
- We prioritized the important emergency-related features first.
- Breaking the User Stories into smaller Subtasks helped us understand what work had to be done for each feature.
- We made good progress on features like ambulance tracking, hospital selection, hospital notification, and sharing patient information.
- The Jira Scrum Board helped us see which tasks were To Do, In Progress, and Done.
- Working with Jira made it easier for our team members to know what tasks were assigned to them.
- We were also able to identify some dependencies between Live Ambulance Tracking and Hospital Coordination while planning the sprint.

3. Challenges Encountered

- Initially, we found it a little difficult to decide how to divide some features into separate User Stories and Subtasks.
- Some features like real-time tracking, ETA calculation, and hospital availability depend on other parts of the system, so planning them took some time.
- We had to discuss how ambulance location information should be shared with the hospital and how the hospital should respond.
- Since the sprint had limited time, we could not include all the planned features in the active sprint.
- Estimating Story Points was also slightly difficult because we were not completely sure about the complexity of some technical tasks.

- We also had to regularly check Jira to make sure that the issue statuses and assignments were updated correctly.

4. Areas for Improvement

- We should decide a clear Sprint Goal before starting the sprint.
- All User Stories and Subtasks should be assigned to team members during the initial sprint planning itself.
- We should identify dependencies between different modules earlier so that they do not affect the progress later.
- We can spend more time on backlog refinement before starting the sprint so that the requirements are clear.
- Components and Versions should be created earlier instead of adding them later.
- We should update the Jira issue status regularly so that the Scrum Board and Burndown Chart show the actual progress.
- We can improve our Story Point estimation by discussing the difficulty of each task together before assigning points.

5. How Scrum Helped

Scrum helped our team organize the project in a better way. Instead of treating the whole Emergency Ambulance Management System as one large task, we divided it into Epics, User Stories, and Subtasks.

The Product Backlog helped us keep track of all the features that had to be developed, while the Sprint Backlog helped us select the features that we wanted to work on during the sprint. The Scrum Board made it easy to track the progress of each task by moving it from To Do to In Progress and then to Done.

Using Jira also helped us divide the work among team members and keep track of what everyone was working on. The Burndown Chart helped us get an idea about how much work was completed and how much was still remaining. Overall, using Scrum gave us a better understanding of how software development can be planned and managed as a team.

CONCLUSION:

Overall, our team made good progress during the sprint and was able to organize the main features of the Emergency Ambulance Management System in Jira. We worked on important areas such as ambulance booking, live tracking, and hospital coordination. We also learned how to create and manage Epics, User Stories, Subtasks, Sprints, and the Scrum Board.

We faced some difficulties while estimating tasks, managing dependencies, and keeping the Jira board updated, but these helped us understand how an actual Agile development process works. In the next sprint, we can improve our planning, task assignment, and communication so that more planned features can be completed within the sprint.